Dovetail Research Proposal

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Red = Areas of significant confusion.
Blue = Comments.

Abstract

This proposal explores whether Touchette and Lloyd's information-theoretic account of control can be extended from a single-step DAG to simple multi-step DAG chains relevant to agent foundations. I outline the original result, identify points of confusion in the conditional analysis and support-based interpretation of closed-loop control, and suggest candidate directions for formalising multi-step entropy reduction. The proposal is primarily exploratory: it aims to clarify the theorem, isolate the key obstacles to generalisation, and develop conjectures and definitions that could support later work on sequential control, information flow, and agent-like intervention.

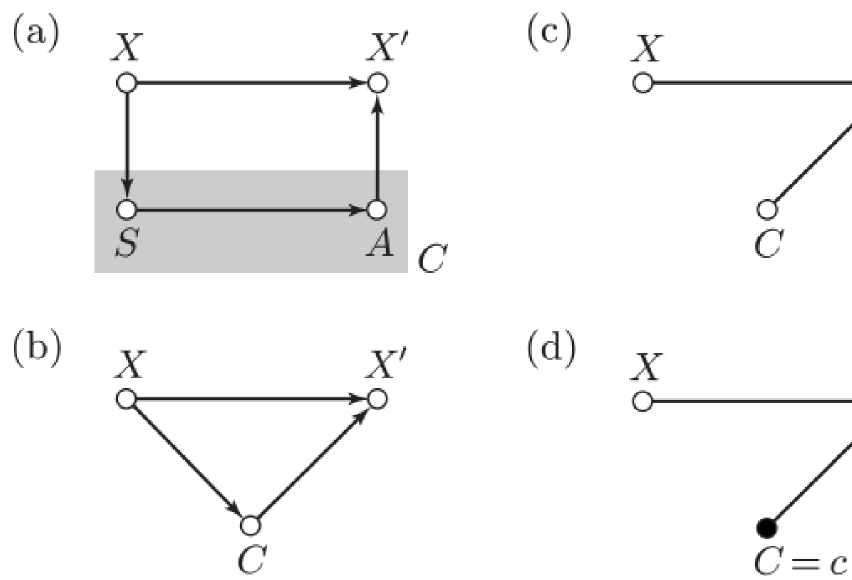

FIG. 1: Directed acyclic graphs representing a basic control process. (a) Full control system with a sensor S and an actuator A. (b) Reduced closed-loop diagram obtained by merging the sensor and the actuator into a single controller device, the controller. (c) Reduced open-loop control diagram. (d) Single actuation channel enacted by the controller's state $C = c$.

Figure 1: Directed acyclic graphs representing the control setup.

1 Problem Notation

1.1 Introducing the Problem via DAGs

Consider Figure 1. Here, each arrow gives a probabilistic dependency among random variables according to the general decomposition,

$$p(x_1, ..x_N) = \Pi_{i=1}^{N} p(x_i|\pi(X_i)). \tag{1}$$

Within this decomposition, we have denoted the set of random variables that are direct parents of X_i as $\pi(X_i)$. In this proposal, all random variables are taken to be discrete random variables. We further note $\pi(X_1) = \emptyset$. In other words, no node is a direct parent of itself.

As is seen represented as a node in Figure 1, we can define and a label an initial state using the discrete random variable X whose values, $x \in \chi$, are drawn according to a fixed probability distribution $p(x)$. Similarly (also seen in Figure 1), we can define a final state using the discrete random variable X' whose values, $x' \in \chi$, are drawn in a manner determined by how the discrete random variable S, referred to as a sensor, and the discrete random variable A, referred to as an actuator, influence the transition from X to X'.

For the purposes of this proof outline, the distinction between sensor and actuator is largely irrelevant and is included only to match the terminology Touchette and Lloyd use when relating their work to the control-theory literature that the proof is intended to help formalise. Moving forward let us consider, as seen in the difference between image 1 a) and image 1 b) within Figure 1, the reduced graph formed by considering the nodes representing random variables A and S as one new node C, referred to as the controller variable, assuming values from some set C of controller actions. This choice is informed owing to the time constraints of this proposal, a more careful distinction between the actuator and sensor roles for is left for future study.

One final thing to briefly note, is that each directed edge can be considered to be time-directed as one way in which we can introduce some physical meaning. Also, I refer to these quantities as discrete random variables throughout. Although, I am not sure if this terminology is exact in every case seen within this Touchette–Lloyd paper.

1.2 Further Notation

We can express the joint distributions describing casual dependencies between states of both the open-loop (see Figure 1c) and the closed-loop (see Figure 1b) control graphs as,

$$p(x, x', c)_{closed} = p_X(x)p(c|x)p(x'|x, c) \tag{2}$$

and,

$$p(x, x', c)_{open} = p(x)p(c)p(x'|x, c). \tag{3}$$

This, just like equation (1), is simply an application of the chain rule,

$$P(A_1, .., A_N|B) = \Pi_{i=1}^{N} P(A_i|A_1.., A_i - 1, B), \tag{4}$$

where each A_i is conditional on some other proposition B (however in this case (and in equation 1) the propositioning on B is redundant). The difference between equations (3) and (2) is that the open case does not contain a casual link between X and C therefore, $\forall c \in C$ and $\forall x \in X$, $p(c|x) = p(c)$.

We refer to $p(c|x)$ as a *direct correlation link*. This correlation can be thought of as a measurement channel, referred to by Touchette and Lloyd as a *sensor measurement channel*. One interesting representation highlighted by Touchette and Lloyd is to note that this correlation allows the controller to gather an amount of information formally quantified by the mutual information,

$$I(X; C) = \sum_{x,c} p(x, c) = log \frac{p(x, c)}{p(x)p(c)} \tag{5}$$

where $p(x, c) := p(x)p(c|x)$. (All logarithms are to base-2 unless stated otherwise). I have not come across work highlighting this particular framing of the mutual information. From an agent foundations perspective, or otherwise, there is perhaps more to consider here for this research.

Furthermore, Touchette and Lloyd state that it is *"from here on convenient to think of the control actions indexed by each value of C as a set of actuation channels, with memoryless transition matrices,*

$$p(x'|x)_c := p(x'|x, c) \tag{6}$$

governing the transmission of the random variable X to X'." For now at least, I am taking this at face value simply a convenient representation. Before turning to the conditional analysis, it is worth noting another feature of Touchette and Lloyd's presentation: they classify the possible actuation sub-dynamics into three cases: one-to-one, many-to-one, and one-to-many. Any attempt I make to extend this proof to DAG chain systems will begin with the simplest of these, namely the one-to-one case. In this setting, for $C = c$, the Shannon entropy associated with the transition matrix in equation (6) is preserved when the sensor measurement channel is noiseless.

With the notation now specified and a few of the representations highlighted by Touchette and Lloyd briefly discussed, we now turn to the conditional analysis they describe using this notation.

2 Conditional Analysis

Using the law of total probability applied over both x and c, we can decompose an open-loop controlled process into an ensemble of actuations each indexed by c as follows,

$$p(x')_{\text{open}} = \sum_x p(x) \left[\sum_c p(x'|x, c)p(c) \right]. \tag{7}$$

It is then noted that this decomposition is equivalent to the overall action of a controller transmitting X through an 'averaged' channel whose transition matrix is given by $p(x'|x)$ (which is equivalent to the sum in the parentheses). The *conditional open-loop entropy reduction* associated with this probability is defined as,

$$\Delta H^c_{\text{open}} := H(X) - H(X'|c)_{\text{open}}, \tag{8}$$

where $H(X'|c) = -\sum_{x'} p(x'|c)log(x'|c)$. Similarly, for the blind case,

$$\Delta H_{\text{open}} := H(X) - H(X')_{\text{open}}. \tag{9}$$

Following the same process as above for the closed-loop case, this time also invoking Bayes' theorem,

$$p(x|c) = \frac{p(c|x)p(x)}{p(c)}, \quad \text{we are arrive at,} \quad p(x')_{\text{open}} = \sum_x p(x) \left[\sum_c p(x'|x, c)p(c) \right], \tag{10}$$

note the corresponding,

$$\Delta H_{\text{closed}} := H(X|c) - H(X'|c)_{\text{closed}}, \quad \text{which holds } \forall c. \tag{11}$$

To conclude this set of results, Touchette and Lloyd note: *"The rationale for decomposing a closed-loop control action into a set of conditional actuations can be justified by observing that a closed-loop controller, after the estimation step, can be thought of as an ensemble of open-loop controllers acting on a set of estimated states."* However, at this stage of my research, I view this conditional analysis as simply a natural way to define and introduce the entropy reductions associated with the decomposed probabilities that make up each system shown in Figure 1.

Now that the relevant entropy reductions for the different system configurations have been introduced, we can proceed to the theorems needed to construct the proof.

3 Relevant Lemmas and Theorems

Lemma 8: For any initial state, X, the open-loop entropy reduction, ΔH_{open}, satisfies,

$$\Delta H_{\text{open}} \leq \Delta H^{C}_{\text{open}}, \tag{12}$$

$$\text{where,} \quad \Delta H_{\text{open}} = \sum_{c} p(c)\Delta H^{c}_{\text{open}} = H(X) - H(X'|C)_{\text{open}}, \quad \text{and,} \quad H^{C}_{\text{open}} = \sum_{c} p(c)\Delta H^{c}_{\text{open}} \tag{13}$$

With equality achieved *iff* $I(X;C) = 0$. The proof of this is as follows: using $H(X') \geq H(X'|C)$, we have,

$$\Delta H_{\text{open}} = H(X) - H(X')_{\text{open}} \leq H(X) - H(X'|C) \implies \Delta H^{c}_{\text{open}} \leq \Delta H_{\text{open}}. \tag{14}$$

Specifically for the equality, if C is statistically independent X' then clearly $H(X'|C) = H(X')$. This implies both that $I(X;C) = 0$ and that $\Delta H_{\text{open}} = \Delta H^{C}_{\text{open}}$. Conversely, $\Delta H_{\text{open}} = \Delta H^{C}_{\text{open}} \implies C$ is independent of X'.

Theorem 9: The entropy reduction achieved by a set of actuation sub-dynamics used in open-loop control is always such that (while Touchette–Lloyd present this result using the language of control theory language, can this be viewed more fundamentally as a a general Shannon entropy information theory result? I think so.),

$$\Delta H(x)_{\text{open}} \leq \max_c \Delta H^{c}_{\text{open}}, \quad \forall p(x). \tag{15}$$

We can prove this as follows: The average conditional entropy is always such that,

$$\min_c H(X'|c) \leq \sum p(c) H(X'|C). \tag{16}$$

This is because the smallest value of some Shannon entropy is necessarily smaller than or equal to the weighted sum of this Shannon entropy. Now, by making use of lemma 8, we may obtain,

$$\Delta H(x)_{\text{open}} \leq \Delta H^{C}_{\text{open}} \leq H(X) - \min H(X'|c) = \max_c \Delta H^{c}_{\text{open}}. \tag{17}$$

Finally, if $C = \hat{c}$ with probability 1, then the two inequalities are saturated since in this case $I(X';C) = 0$ and $\Delta H^{c}_{\text{open}} = \Delta H^{\hat{c}}_{\text{open}}$.

Theorem 10: The amount of entropy reduction,

$$\Delta H_{\text{closed}} = H(X) - H(X')_{\text{closed}}, \tag{18}$$

that can be extracted from a system with a given initial state X, by using a closed-loop controller with a fixed set of actuation subdynamics (Again, I believe this just means a fixed set of discrete variables, c, that C can take?), satisfies,

$$\Delta H_{\text{closed}} \leq \Delta^{max}_{open} + I(X;C), \quad \text{where,} \quad \Delta H^{max}_{open} = \max_{p(x),c} \Delta H^{c}_{\text{open}}, \tag{19}$$

represents the maximum entropy decrease that can be obtained over any input distribution. This is proved as follows: given that ΔH^{max}_{open} is defined as the optimal entropy reduction for open-loop control over any distribution, we can write $H(X') \geq H(X) - H(x)^{max}_{open}$. Now, alongside this, *"using the fact that a closed-loop controller is formally equivalent to an ensemble of open-loop controllers acting on the conditional supports* $\text{Supp}(X|c)$ *instead of* $\text{Supp}(X)$" (For now, I'm taking this as given, however this requires looking into more rigorously), we also have, $\forall c$,

$$H(X'|c)_{\text{closed}} \geq H(X) - \Delta H^{\max}_{\text{open}}, \quad \text{and, on average,} \quad H(X'|C) \geq H(X|C) - \Delta H^{\max}_{\text{open}}. \tag{20}$$

This (I believe?) implies that $\Delta H^{\max}_{\text{open}}$ must enter the lower bounds of $H(X')_{\text{open}}$ and $H(X')_{\text{closed}}$. As it stands, this is confusing me a bit. To overcome this I need to revisit the end of the conditional analysis section.

Now, note $H(X') \geq H(X'|C)$ implies,

$$\Delta H(X')_{closed} \geq H(X|C) - \Delta H^{max}_{open}, \tag{21}$$

therefore, we obtain,

$$H_{\text{closed}} \leq H(X) - H(X|C) +^{max}_{closed} = I(X;C) + H^{max}_{closed} \tag{22}$$

Which is the desired upper bound. Finally, to close the proof, Touchette and Lloyd note that H^{max}_{closed} cannot be evaluated using $p(x)$ alone because the maximum reduction of entropy in open-loop control starting from $p(x)$ may differ from the reduction of entropy obtained when some actuation channel is applied in closed-loop to $p(x|c)$. They then provide a specific example to demonstrate this. (I have not yet examined this example and this would be a step to take shortly.)

4 Remarks

The preceding section contains a preliminary outline of the Touchette–Lloyd theorem, stated in equation (22). My initial impression is that, as with many well-written proofs, the result feels natural once the argument is followed carefully. Its manipulations (I think?) depend only on standard results from probability and Shannon information theory; Bayes' theorem, the law of total probability, the chain rule, and the definition of entropy as examples. My working hypothesis is that the proof can be extended from the three-node DAG case to a DAG chain (see Figure 2) self contained to this pretty standard Shannon information theory framework.

Figure 2: Figure showing the DAG chain equivalent of the system this proof outline is concerned with. This proposal is to lay the groundwork for extending the Touchette-Lloyd theorem (equation (22)) to this multi-step setup.

My difficulties were less mathematical than editorial: deciding how much of the paper needed to be included to make this proof outline coherent. Touchette and Lloyd are rigorous, but they also take care to explain what intermediate expressions represent physically. For instance, by interpreting a conditional probability as a transition matrix (their equation 5), or by reordering sums to highlight different but functionally equivalent perspectives on the same distribution (their equations 8 and 10). I am not yet confident that I fully appreciate the role of all these perspectives. For now, the best way forward seems to be to begin working on novel DAG-chain proofs and then return to this outline once the demands of producing a new result clarify which of its omissions matter.

For the immediate task of extending this proof to multiple DAG chains, the relevant literature appears largely self-contained within this paper. My next step is to return to the end of the conditional analysis section and work carefully through the role of the supports. As supplementary reading, Touchette and Lloyd frequently cite H. Touchette and S. Lloyd, Phys. Rev. Lett. 84, 1156 (2000), which offers alternative proofs of some results and develops related ideas further. This may prove useful if a direct line-by-line extension from the three-node DAG case to a DAG chain meets an obstacle, since it could point to a theorem that is easier to generalise. It may also be helpful to extend some of the paper's examples to DAG-chain settings, in order to build intuition in less abstract cases. Finally, it may be worth exploring the time-dependent case to see whether discretised time steps can be treated as DAG nodes, thereby linking the dynamical and graphical viewpoints. Doing so would likely require reading more broadly in adjacent information-theoretic literature.

As a first conjectural step, now that this outline is in place, I would extend the system by adding an additional C_2 and X_3, giving the simplest multi-step case shown in Figure 2 without any further nodes. I would begin with the noiseless one-to-one discrete setting, assuming that each variable in $\{X_1, X_2, X_3, C_1, C_2\}$ takes values in a finite set of common size N, so that $|X_i| = |C_i| = N$ for all relevant i.

Finally, my apologies for not including such a conjecture in this proposal. In retrospect, while this condensed proof outline has been useful, it likely took more time than ideal, and a shorter version might have allowed me to begin exploring preliminary conjectures sooner. Please do not hesitate to get in touch if it would be useful for me to do so between Tuesday the 26th and Thursday the 28th.